VidiFlow LEX — Hugging Face Datasets Study Report

Date: 2026-09-14 · Source: Hugging Face Hub live screenshots · Purpose: pick best tech + AI term dataset for LEX (Directus `lexi_terms` on new server, CDN `cdn.vidi.news` → R2 bucket `vidismart`)

Recommendation (hybrid — no single set matches LEX schema): seed term list from **AI-Dictionary (16.6k, MIT) ∩ GIST (5k, Apache-2.0)** → acronym expansions from **arxiv-acronym-gen (8k)** → definitions + related terms from **OpenGloss (506k, CC-BY-4.0, reading_level + domain tags)** filtered to technology. Keep LEX editorial layer (oneLiner, anatomy, flow, brands, smartStack, video) in Directus. Use PLOD / acronym_identification only to train hover-detection, never as shipped content.

Comparison table

#	Dataset	Rows	License	Fields	LEX fit
1	J0nasW/AI-Dictionary	16,275	MIT	keyword, weight, embedding	Best AI seed list; noisy, no defs
2	Jerry999/multilingual-terminology (GIST)	~5,000 EN	Apache-2.0	English + AR/ZH/FR/JA/RU	Cleanest AI terms, research-grade
3	sjmoran/arxiv-acronym-gen	8,070	Apache-2.0	acronym, title	Acronym format ideal; science-skewed
4	mjbommar/opengloss-v1.1-definitions	506k	CC-BY-4.0	definition, synonyms, hypernyms, examples, reading_level, domain tags	Closest to LEX one-liner + related
5	amirveyseh/acronym_identification	17.5k sents	research	tokens, BIO labels	Detection model only
6	surrey-nlp/PLOD-filtered	160k+ segs	research	tokens, pos, ner AC/LF	Detection model only
7	aheadmint/machine-learning-glossary-ai	small curated	check	term, math, MLOps refs	Curated ML/RAG reference
8	cdiamond/ai-infrastructure-economics-glossary	11	CC-BY-4.0	term, definition, coined_by	Editorial example only
9	conceptnet5/conceptnet5	millions	CC-BY-4.0 mix	arg1, rel, arg2, weight	Related-terms graph
10	jake-anto/wiktionary	10.6M entries	CC-BY-SA	entries/senses/examples/translations/sounds/linkages	General fallback defs
11	brennercruvinel/3char	751k codes	mixed/authority	system, authority, regex, codes	Infra abbreviations (HTTP/CSS/MIME)
12	cstr/en-wiktionary-sqlite-all	1.4M entries	CC-BY-SA	SQLite 40+ tables, senses, wikilinks	Offline/mobile-ready lexicon

1 — J0nasW/AI-Dictionary (16.6k AI phrases, MIT)

Whole-AI-domain phrase list with weights + embeddings. Best seed coverage; needs dedupe (e.g. duplicates like "graph neural networkand") and has no definitions — join to OpenGloss/Wiktionary for glosses.

Datasets: J0nasW/AI-Dictionary

Like 0

Tasks: Text Classification | Feature Extraction
Modalities: Text
Formats: csv
Languages: English
Size: 10K - 100K
Tags: code
Libraries: Datasets | pandas | Croissant | +1
License: mit

Dataset Viewer — Auto-converted to Parquet | API | Embed | Duplicate | Data Studio

Split (1)
train · 16.3k rows

Search this dataset

keyword string · *lengths* 4 – 52	weight float64 0 – 1	embedding string · *lengths* 15.8k – 16k
graph neural network	1	[0.08722702413797379, 0.8745535016059875, -0.3408846855163574, -0.0734473466873169,…
graph neuralnetworks	1	[-0.11062880605459213, 0.9918195605278015, -0.2675487995147705, -0.16344797611236572,…
graph neural networkand	1	[0.1356668621301651, 0.8163174986839294, -0.598024249076843, -0.025542443618178368,…
graph neural networks based	1	[-0.10963953286409378, 0.6333564519882202, -0.4167910516262054, -0.1791401356458664,…
graph neural network based	1	[-0.029546916484832764, 0.5460565686225891, -0.3735005855560303, -0.11402704566717148,…
neural network	0.990941	[-0.02977747470140457, 0.10993075370788574, -0.21777033805847168, -0.6170248985290527,…

‹ Previous | 1 | 2 | 3 | ... | 163 | Next ›

→ Copy to bucket NEW | Use this dataset

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Number of rows: 16,275
Total file size: 260 MB

AI Dictionary Dataset

Welcome to the AI Dictionary dataset on HuggingFace. This dataset is a comprehensive tool comprised of 16,665 unique key phrases that describe the whole domain of Artificial Intelligence (AI). It serves both the research community and industry domains, aiding in the identification of radical innovations and uncovering applications of AI in new domains.

This dataset is the result of the research paper "The AI Dictionary: The Foundation for a Text-Based Tool to Identify and Measure Technology Innovation". The paper explores the rapidly evolving landscape of AI as a General Purpose Technology and its dual role in driving and sustaining innovation across various domains.

The AI Dictionary is designed to measure technological innovation using text-based methods. We hope to establish a foundational methodology for a new and innovation measurement tool.

Related Repository

The process of creating and validating the AI Dictionary is detailed in a series of Jupyter notebooks and Python scripts in the related GitHub repository. You can access the repository at the following link:

AI Dictionary GitHub Repository

Please refer to the repository for a deeper understanding of the methodology and the process behind the creation of the AI Dictionary.

Usage

You can use this dataset for a variety of AI and Machine Learning tasks such as text classification, named entity recognition, and more. The dataset can also be used for research purposes to identify and measure technological innovation in the field of AI.

License

This dataset is released under the MIT License.

2 — GIST multilingual-terminology (~5k AI terms, Apache-2.0)

Sourced from award papers across 18 top AI conferences, LLM-extracted + human-translated + GPT-4o-validated. Highest-quality AI term list; viewer currently broken (CSV per language); terms only.

Datasets: Jerry999/multilingual-terminology

Like 1

Languages: English, Arabic, Chinese +3 | Size: 1K<n<10K | ArXiv: arxiv:2412.18367 | Tags: ai, terminology, multilingual, translation, nlp, scientific-text

License: apache-2.0

Dataset Preview API Embed Duplicate Data Studio

Split (5)
Arabic

▶ The full dataset viewer is not available (click to read why). Only showing a preview of the rows.

Index int64	English string	Arabic string
0	10-fold cross validation	التحقق المتقاطع بعشر طيات
1	1D convolution	الالتفاف أحادي البُعد
2	2 norm	النورم 2
3	2D convolution	الالتفاف ثنائي الأبعاد
4	2D image	صورة ثنائية الأبعاد
5	2D image synthesis	توليف الصور ثنائية الأبعاد
6	2D-3D correspondence	المراسلات ثنائية وثلاثية الأبعاد
7	3D bounding box	الصندوق المحيط ثلاثي الأبعاد
8	3D computer vision	الرؤية الحاسوبية ثلاثية الأبعاد
9	3D convolutional network	شبكة تلافيفية ثلاثية الأبعاد

GIST: Glossary of Multilingual AI Scientific Terminology

Paper Title: **Towards Global AI Inclusivity: A Large-Scale Multilingual Terminology Dataset (GIST)**

Website Demo Instructions: https://github.com/jiarui-liu/MultilingualAITerminology

Dataset Summary

GIST is a large-scale, high-quality multilingual AI terminology dataset developed to support global inclusivity in AI research. It consists of around **5,000 English AI-specific terms**, each translated into **Arabic, Chinese, French, Japanese, and Russian**.

Terms are sourced from award-winning papers across 18 top-tier AI conferences (2000–2023). The dataset was created using a hybrid pipeline that combines:

- LLM-based term extraction and filtering
- Expert human translation via crowdsourcing
- GPT-4o-based validation and refinement

Citation

If you use this dataset, please cite:

```
@article{liu2024towards,
  title={Towards Global AI Inclusivity: A Large-Scale Multilingual Terminology Dataset},
  author={Liu, Jiarui and Ouzzani, Iman and Li, Wenkai and Zhang, Lechen and Ou, Tianyue and I
  journal={arXiv preprint arXiv:2412.18367},
  year={2024}
}
```

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Total file size: 1.34 MB

Paper for Jerry999/multilingual-terminology

Towards Global AI Inclusivity: A Large-Scale Multi…
Paper • 2412.18367 • Published Dec 24, 2024

3 — arxiv-acronym-gen (8,070 acronym→title)

Compact JSON (742kB). Ideal acronym→expansion shape for LEX, but arXiv science skew (MICE, LIGO) — little enterprise tech (CRM/SaaS/KPI). Use as expansion pattern + retrieval eval.

Datasets: sjmoran / arxiv-acronym-gen

Like 0

Modalities: Text | Formats: json | Languages: English | Size: 1K - 10K

Tags: acronym | text-generation | scientific-writing | arxiv | natural-language-processing | abbreviation | Libraries: Datasets | pandas | Croissant + 1

License: apache-2.0

Dataset Viewer — Auto-converted to Parquet | API | Embed | Duplicate | Data Studio

Split (1)
train · 8.07k rows

Search this dataset

acronym string · lengths	title string · lengths
ROPS	Real Options for Project Schedules
ECHO	an Eulerian Conservative High Order scheme for general
QMC	A Model Checker for Quantum Systems
DWEB	A Data Warehouse Engineering Benchmark
DOEF	A Dynamic Object Evaluation Framework
BLOX	The Bonn Lensing, Optical, and X-ray selected galaxy clusters --
VSOP	The Variable Star One-shot Project I. Project presentation and
GDPS	The Gemini Deep Planet Survey -
MICE	The International Muon Ionization Cooling Experiment

< Previous 1 2 3 ... 81 Next >

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Number of rows: 8,070 | Total file size: 742 kB

AcronymGen-Titles 🧠

AcronymGen-Titles is a dataset of over 8,000 high-quality acronym–title pairs extracted from scientific papers on **arXiv**. Each entry contains a scientific acronym and the corresponding paper title in which it appears. This variant contains **title-only pairs** — no abstracts — offering a compact and focused dataset for acronym generation, matching, or compression tasks.

📊 Dataset Overview

- 📦 **Size**: 8,070 examples
- 🧠 **Source**: Extracted from arXiv.org
- 🔍 **Fields**:
  - `acronym`: Scientific acronym used in the paper (e.g., "MICE", "GCP")
  - `title`: Title of the paper containing or describing the acronym

---

✨ Example

```
{
  "acronym": "MICE",
  "title": "The International Muon Ionization Cooling Experiment"
}
```

This dataset is ideal for training or evaluating acronym suggestion models, particularly where only the title context is available. It can also be used for retrieval, embedding compression, or understanding domain-specific abbreviation patterns.

4 — OpenGloss v1.1 definitions (506k senses, CC-BY-4.0)

Synthetic encyclopedic dictionary: definition + synonyms/antonyms/hypernyms/hyponyms + collocations + examples + reading_level (K–PhD) + domain tags. Filter domain:technology for LEX one-liner drafts and connectsWith graph. Requires attribution.

Datasets: mjbommar/**opengloss-v1.1-definitions** | Like 1

Tasks: Text Generation | Question Answering | Text Classification +1 | Modalities: Tabular | Text | Formats: json | Languages: English | Size: 100K - 1M

Tags: dictionary | lexicon | wordnet | semantic-network | knowledge-graph | encyclopedic +3 | Libraries: Datasets | pandas | Croissant +1 | License: cc-by-4.0

Dataset Viewer — Auto-converted to Parquet | API | Embed | Duplicate | Data Studio

Split (1)
train · 506k rows

Search this dataset

id string · *lengths* 8 – 55	word string · *lengths* 1 – 43	part_of_speech string · *classes* 18 values	sense_index int64 0 – 4	global_sense_inde int64 0 – 2
complect_verb_0	complect	verb	0	
complect_verb_1	complect	verb	1	
waterborne_adjective_0	waterborne	adjective	0	
waterborne_adjective_1	waterborne	adjective	1	
waterborne_adjective_2	waterborne	adjective	2	
physical_condition_noun_0	physical condition	noun	0	

< Previous 1 2 3 ... 5,062 Next >

→ Copy to bucket NEW | </> Use this dataset

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Number of rows: 506,143
Total file size: 2.43 GB

Collection including mjbommar/opengloss-v1.1-…

OpenGloss 1.x Collection
OpenGloss v1.1–v1.3: 206K LLM-gen… • 13 items • Up…

OpenGloss Dictionary v1.1 (Definition-Level)

Dataset Summary

OpenGloss is a synthetic encyclopedic dictionary and semantic knowledge graph for English that integrates lexicographic definitions, encyclopedic context, etymological histories, and semantic relationships in a unified resource.

This dataset provides the **definitions-level view** where each record represents one sense definition.

Key Statistics

- **506,143 sense definitions**
- **7,701,312 semantic edges** (synonyms, antonyms, hypernyms, hyponyms, collocations, inflections)
- **150,637 entries** with encyclopedic content (100.0% coverage)
- **150,637 entries** with etymology (100.0% coverage)
- **149,734 entries** with Wikipedia frequency data (99.4% coverage)
- **100% reading level coverage** (K through PhD scale)
- **100% domain tag coverage** (10+ subject domain categories)
- **Average 3.36 senses per lexeme**
- **Average 51.1 edges per lexeme**

What's New in v1.1?

Compared to OpenGloss v1.0 (150,101 words):

1. **Wikipedia Frequency Data** (~99.5% coverage): Raw occurrence counts and frequency ranks from Wikipedia, enabling importance-based filtering and sorting
2. **Educational Reading Levels** (100% coverage): Lexemes tagged with grade levels (K, 1-12, BS, PhD) for curriculum alignment and differentiated instruction
3. **Domain Tags** (100% coverage): Subject-area classification (language, science, technology, society, history, etc.) for content-specific applications
4. **Structured Etymology Segments**: Detailed historical trail with language, era, gloss, and citation sources (enhanced from v1.0's text summaries)
5. **Structured Morphology**: 7 inflection types (plural, past_tense, comparative, etc.) + 4 derivation types per POS
6. **Rich Edge Metadata**: Semantic relationships now include domain context and educational features

7. **Hierarchical POS Entries**: New structured format alongside backward-compatible flattened senses
8. **Audit Timestamps**: created_at/updated_at fields for version tracking and data provenance

POS Distribution

Part of Speech	Count
noun	122,564
adjective	55,905
verb	36,420
adverb	5,583
determiner	1,510
preposition	1,234
interjection	938
pronoun	395
conjunction	249
particle	18
proper noun	9
numeral	5
proper_noun	4
prefix	2
suffix	1
abbreviation	1
adjetivo	1
sustantivo	1

Edge Type Distribution

Relationship Type	Count
synonym	1,485,130
hyponym	1,285,557
collocation	1,227,353
antonym	1,007,039
hypernym	994,981
etymology_parent	697,485
inflection	353,210
derivation_noun	279,501
derivation_adjective	163,035
derivation_verb	119,944
derivation_adverb	68,105
cognate	19,972

Loading the Dataset

```
from datasets import load_dataset

# Load the full dataset
dataset = load_dataset("mjbommar/opengloss-v1.1-definitions")

# Access records
for record in dataset["train"]:
    print(f"Word: {record['word']}")
    print(f"Senses: {record['total_senses']}")
```

```
    print(f"Edges: {record['total_edges']}\n")
```

New v1.1 Fields & Usage Examples

Wikipedia Frequency Data

Filter by word importance using frequency data:

```
# Get high-frequency words (top 10,000)
common_words = dataset["train"].filter(
    lambda x: x["wiki_frequency_rank"] is not None and x["wiki_frequency_rank"] <= 10000
)

# Sort by frequency
sorted_by_freq = dataset["train"].sort("wiki_frequency", reverse=True)
```

Reading Levels

Filter vocabulary by grade level for educational applications:

```
# Elementary (K-5)
elementary = dataset["train"].filter(lambda x: x["reading_level"] in ["K", "1", "2", "3", "4"

# Middle school (6-8)
middle_school = dataset["train"].filter(lambda x: x["reading_level"] in ["6", "7", "8"])

# High school (9-12)
high_school = dataset["train"].filter(lambda x: x["reading_level"] in ["9", "10", "11", "12"]

# Advanced (BS/PhD)
advanced = dataset["train"].filter(lambda x: x["reading_level"] in ["BS", "PhD"])
```

Domain Tags

Filter by subject area for content-specific applications:

```
# Science vocabulary
science_words = dataset["train"].filter(
    lambda x: any("science" in tag or "life-sciences" in tag for tag in x.get("tags", []))
)

# Technology vocabulary
tech_words = dataset["train"].filter(
    lambda x: any("technology" in tag for tag in x.get("tags", []))
)

# Social studies
social_studies = dataset["train"].filter(
    lambda x: any(tag.startswith("domain:history") or tag.startswith("domain:society")
                  for tag in x.get("tags", []))
)
```

Etymology Segments

Access structured etymology with language trail:

```
# Words with detailed etymology
words_with_etymology = dataset["train"].filter(lambda x: len(x.get("etymology_segments", []))

# Find words from specific language origins
latin_origin = dataset["train"].filter(
    lambda x: any(seg.get("language", "").lower() == "latin"
                  for seg in x.get("etymology_segments", []))
)
```

Citation

If you use OpenGloss in your research, please cite:

```
@misc{bommarito2025opengloss,
  title={OpenGloss: A Synthetic Encyclopedic Dictionary and Semantic Knowledge Graph},
  author={Bommarito, Michael J., II},
  year={2025},
```

```
  url={https://huggingface.co/datasets/mjbommar/opengloss-v1.1-definitions},
  note={Dataset available under CC-BY 4.0}
}
```

License

This dataset is released under **Creative Commons Attribution 4.0 International (CC-BY 4.0)**.

Version History

- **v1.1** (2025-11): Enhanced release with structured morphology, etymology segments, and frequency data
- **v1.0** (2025-01): Initial release

Acknowledgments

This dataset was generated using:

- pydantic-ai for structured LLM generation
- OpenAI GPT models for content generation
- Anthropic Claude for quality assurance

Generated from the OpenGloss v1.1 dataset.

System theme TOS Privacy About Careers Models Datasets Spaces Pricing Docs

5 — acronym_identification (AAAI-21 SDU)

17.5k manually annotated sentences, BIO scheme (B-short/I-short/B-long/I-long). For training the hover detector (which tokens are acronyms), not for shipped definitions.

Datasets: amirveyseh/acronym_identification

Like 23

Tasks: Token Classification | Modalities: Text | Formats: parquet | Languages: English | Size: 10K - 100K | ArXiv: arxiv:2010.14678 | Tags: acronym-identification

Libraries: Datasets | pandas | Croissant | +1 | License: mit

Dataset Viewer

Auto-converted to Parquet | API | Embed | Duplicate | Data Studio

Split (3)
train · 14k rows

Search this dataset

id string · lengths	tokens sequence · lengths	labels sequence · lengths
TR-0	["What", "is", "here", "called", "controlled", "natural", "language", "…	[4, 4, 4, 4, 0, 2, 2, 4, 1, 4, 4, 4, 4, 4, 4, 4, 4]
TR-1	["In", "this", "paper", ",", "we", "introduce", "CoQA", ",", "a",…	[4, 4, 4, 4, 4, 4, 1, 4, 4, 0, 2, 2, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4,…
TR-2	["in", ":", "Proceedings", "of", "SODA", "98", ",", "Ninth", "Annual",…	[4, 4, 4, 4, 1, 4, 4, 4, 4, 1, 4, 1, 4, 4, 4, 4, 4, 4, 4, 4, 1, 4, 1, 4, 4,…
TR-3	["But", "is", "the", "existing", "HPC", "I", "/", "O", "stack", "still",…	[4, 4, 4, 4, 1, 3, 3, 3, 4, 4, 4, 4, 4, 1, 4, 4, 4, 4, 4]
TR-4	["For", "a", "fixed", "and", "known", ",", "we", "apply", "a", "successive",…	[4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 0, 2, 2, 4, 1, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4,…
TR-5	["Section", "II", "introduces", "issues", "relevant", "to", "VMP", "."]	[4, 4, 4, 4, 4, 4, 1, 4]

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Dataset Card for Acronym Identification Dataset

Dataset Summary

This dataset contains the training, validation, and test data for the **Shared Task 1: Acronym Identification** of the AAAI-21 Workshop on Scientific Document Understanding.

Supported Tasks and Leaderboards

The dataset supports an `acronym-identification` task, where the aim is to predic which tokens in a pre-tokenized sentence correspond to acronyms. The dataset was released for a Shared Task which supported a leaderboard.

Languages

The sentences in the dataset are in English (`en`).

Dataset Structure

Data Instances

A sample from the training set is provided below:

```
{'id': 'TR-0',
 'labels': [4, 4, 4, 4, 0, 2, 2, 4, 1, 4, 4, 4, 4, 4, 4, 4, 4, 4],
 'tokens': ['What',
  'is',
  'here',
  'called',
  'controlled',
  'natural',
  'language',
  '(',
  'CNL',
  ')',
  'has',
  'traditionally',
  'been',
  'given',
  'many',
  'different',
  'names',
```

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Downloads last month 54,890

Homepage: sites.google.com
Repository: github.com
Paper: What Does This Acronym Mean? Introducing a New Dat…
Leaderboard: competitions.codalab.org
Number of rows: 17,473
Total file size: 2.08 MB

Models trained or fine-tuned on amirveyseh/acr…

- tomaarsen/span-marker-bert-base-uncas… — Token Classification · 0.1B · Up… · 240k · 6
- sileod/deberta-v3-base-tasksource-nli — Zero-Shot Classification · 0.2B · 21.8k
- sileod/deberta-v3-large-tasksource-nli — Zero-Shot Classification · 0.4B · 3.35k
- tomaarsen/span-marker-bert-base-acron… — Token Classification · 0.1B · Upda… · 80 · 7
- sileod/deberta-v3-xsmall-tasksource-n… — Zero-Shot Classification · Updated J… · 77 · 3
- lewtun/autotrain-acronym-identificati… — Token Classification · Updated Aug 25, 2022 · 49

Space using amirveyseh/acronym_identific… 1

rdassignies/dataset_card_evaluator

Paper for amirveyseh/acronym_identification

What Does This Acronym Mean? Introducing a Ne…
Paper · 2010.14678 · Published Oct 28, 2020 · 1

```
    '.']}
```

Please note that in test set sentences only the `id` and `tokens` fields are available. `labels` can be ignored for test set. Labels in the test set are all `O`

Data Fields

The data instances have the following fields:

- `id`: a `string` variable representing the example id, unique across the full dataset
- `tokens`: a list of `string` variables representing the word-tokenized sentence
- `labels`: a list of `categorical` variables with possible values `["B-long", "B-short", "I-long", "I-short", "O"]` corresponding to a BIO scheme. `-long` corresponds to the expanded acronym, such as *controlled natural language* here, and `-short` to the abbrviation, `CNL` here.

Data Splits

The training, validation, and test set contain `14,006`, `1,717`, and `1750` sentences respectively.

Dataset Creation

Curation Rationale

> *"First, most of the existing datasets for acronym identification (AI) are either limited in their sizes or created using simple rule-based methods. This is unfortunate as rules are in general not able to capture all the diverse forms to express acronyms and their long forms in text. Second, most of the existing datasets are in the medical domain, ignoring the challenges in other scientific domains. In order to address these limitations this paper introduces two new datasets for Acronym Identification. Notably, our datasets are annotated by human to achieve high quality and have substantially larger numbers of examples than the existing AI datasets in the non-medical domain. "*

Source Data

Initial Data Collection and Normalization

> *"In order to prepare a corpus for acronym annotation, we collect a corpus of 6,786 English papers from arXiv. These papers consist of 2,031,592 sentences that would be used for data annotation for AI in this work."*

The dataset paper does not report the exact tokenization method.

Who are the source language producers?

The language was comes from papers hosted on the online digital archive arXiv. No more information is available on the selection process or identity of the writers.

Annotations

Annotation process

> *"Each sentence for annotation needs to contain at least one word in which more than half of the characters in are capital letters (i.e., acronym candidates). Afterward, we search for a sub-sequence of words in which the concatenation of the first one, two or three characters of the words (in the order of the words in the sub-sequence could form an acronym candidate. We call the sub-sequence a long form candidate. If we cannot find any long form candidate, we remove the sentence. Using this process, we end up with 17,506 sentences to be annotated manually by the annotators from Amazon Mechanical Turk (MTurk). In particular, we create a HIT for each sentence and ask the workers to annotate the short forms and the long forms in the sentence. In case of disagreements, if two out of three workers agree on an annotation, we use majority voting to decide the correct annotation. Otherwise, a fourth annotator is hired to resolve the conflict"*

Who are the annotators?

Workers were recruited through Amazon MEchanical Turk and paid $0.05 per annotation. No further demographic information is provided.

Personal and Sensitive Information

Papers published on arXiv are unlikely to contain much personal information, although some do include some poorly chosen examples revealing personal details, so the data should be used with care.

Considerations for Using the Data

Social Impact of Dataset

[More Information Needed]

Discussion of Biases

[More Information Needed]

Other Known Limitations

Dataset provided for research purposes only. Please check dataset license for additional information.

Additional Information

Dataset Curators

[More Information Needed]

Licensing Information

The dataset provided for this shared task is licensed under CC BY-NC-SA 4.0 international license.

Citation Information

```
@inproceedings{Veyseh2020,
  author    = {Amir Pouran Ben Veyseh and
               Franck Dernoncourt and
               Quan Hung Tran and
               Thien Huu Nguyen},
  editor    = {Donia Scott and
               N{\'{u}}ria Bel and
               Chengqing Zong},
  title     = {What Does This Acronym Mean? Introducing a New Dataset for Acronym
               Identification and Disambiguation},
  booktitle = {Proceedings of the 28th International Conference on Computational
               Linguistics, {COLING} 2020, Barcelona, Spain (Online), December 8-13,
               2020},
  pages     = {3285--3301},
  publisher = {International Committee on Computational Linguistics},
  year      = {2020},
  url       = {https://doi.org/10.18653/v1/2020.coling-main.292},
  doi       = {10.18653/v1/2020.coling-main.292}
}
```

Contributions

Thanks to @abhishekkrthakur for adding this dataset.

System theme TOS Privacy About Careers Models Datasets Spaces Pricing Docs

6 — PLOD-filtered (160k+ abbreviation segments)

PLOS-journal abbreviation detection with RoBERTa/AlBERT baselines (F1 ~0.92). Same use as #5: detection model training.

Datasets: surrey-nlp / PLOD-filtered | Like 1 | Follow NLP-AI Group @ Univer... 159

Tasks: Token Classification | Modalities: Text | Languages: English | Size: 100K - 1M | ArXiv: arxiv:2204.12061 | Tags: abbreviation-detection

Libraries: Datasets | Croissant | License: cc-by-sa-4.0

Dataset Viewer — Auto-converted to Parquet | API | Embed | Duplicate | Data Studio

Split (3): train · 113k rows

Search this dataset

id string · lengths	tokens sequence	pos_tags sequence	ner_tags sequence
0	["Alternatively", ",", "fibroblasts", "were",…	[2, 13, 8, 3, 16, 2, 14, 14, 11, 3,…	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0…
1	["Study", "-", "specific", "risk", "ratios", "(", "RRs",…	[8, 13, 0, 8, 8, 13, 12, 13, 5, 0,…	[0, 0, 0, 3, 4, 0, 1, 0, 0, 0, 1, 0, 0…
2	["Three", "faba", "bean", "genotypes", ",", "Aguadulce",…	[9, 8, 8, 8, 13, 12, 13, 12, 13, 13,…	[0, 0, 0, 0, 0, 3, 0, 1, 0, 0, 0, 0, 0…
3	["Neurological", "complications", "of",…	[0, 8, 1, 8, 16, 0, 8, 8, 5, 8, 13, 0,…	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0…
4	["Reference", "values", "(", "RVs", ")", "of", "each",…	[8, 8, 13, 12, 13, 1, 6, 8, 3, 16, 1,…	[3, 4, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0…
5	["We", "next", "undertook", "a", "large", "case", "study",…	[11, 2, 16, 6, 0, 8, 8, 1, 6, 9, 8,…	[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0…

< Previous 1 2 3 ... 1,127 Next >

PLOD: An Abbreviation Detection Dataset

This is the repository for PLOD Dataset published at LREC 2022. The dataset can help build sequence labelling models for the task Abbreviation Detection.

Dataset

We provide two variants of our dataset - Filtered and Unfiltered. They are described in our paper here.

1. The Filtered version can be accessed via Huggingface Datasets here and a CONLL format is present here.
2. The Unfiltered version can be accessed via Huggingface Datasets here and a CONLL format is present here.
3. The SDU Shared Task data we use for zero-shot testing is available here.

Dataset Card for PLOD-filtered

Dataset Summary

This PLOD Dataset is an English-language dataset of abbreviations and their long-forms tagged in text. The dataset has been collected for research from the PLOS journals indexing of abbreviations and long-forms in the text. This dataset was created to support the Natural Language Processing task of abbreviation detection and covers the scientific domain.

Supported Tasks and Leaderboards

This dataset primarily supports the Abbreviation Detection Task. It has also been tested on a train+dev split provided by the Acronym Detection Shared Task organized as a part of the Scientific Document Understanding (SDU) workshop at AAAI 2022.

Languages

English

Dataset Structure

Data Instances

A typical data point comprises an ID, a set of `tokens` present in the text, a set of `pos_tags` for the corresponding tokens obtained via Spacy NER, and a set of `ner_tags` which are limited to `AC` for `Acronym` and `LF` for `long-forms`.

→ Copy to bucket NEW | Use this dataset

Downloads last month 105

Repository: github.com | Paper: arxiv.org | Leaderboard: paperswithcode.com

Point of Contact: Diptesh Kanojia | Number of rows: 160,932 | Total file size: 121 MB

Models trained or fine-tuned on surrey-nlp/PLO...

surrey-nlp/roberta-base-finetuned-abbr
Token Classification • Updated Dec ... • 145 • 1

kaejo98/acronym-definition-detection
Token Classification • 0.3B • Updated Se... • 12

ArshiaKarimian/CW_NER_MODEL_PIPELINE
Token Classification • 66.4M • Updated ... • 11

EngTig/roberta-large-finetuned-ner-fi...
Token Classification • 0.4B • Updated Ap... • 10

cccmatthew/surrey-gp30
Token Classification • Updated May 23, 2024 • 1

Spaces using surrey-nlp/PLOD-filtered 2

Sevixdd/Sevixdd-roberta-base-finetuned-ner

lobrien001/Lee-Sevixdd-roberta-base-finetuned-ner

Paper for surrey-nlp/PLOD-filtered

PLOD: An Abbreviation Detection Dataset for Scie...
Paper • 2204.12061 • Published Apr 26, 2022 • 1

An example from the dataset: {'id': '1', 'tokens': ['Study', '-', 'specific', 'risk', 'ratios', '(', 'RRs', ')', 'and', 'mean', 'BW', 'differences', 'were', 'calculated', 'using', 'linear', 'and', 'log', '-', 'binomial', 'regression', 'models', 'controlling', 'for', 'confounding', 'using', 'inverse', 'probability', 'of', 'treatment', 'weights', '(', 'IPTW', ')', 'truncated', 'at', 'the', '1st', 'and', '99th', 'percentiles', '.'], 'pos_tags': [8, 13, 0, 8, 8, 13, 12, 13, 5, 0, 12, 8, 3, 16, 16, 0, 5, 0, 13, 0, 8, 8, 16, 1, 8, 16, 0, 8, 1, 8, 8, 13, 12, 13, 16, 1, 6, 0, 5, 0, 8, 13], 'ner_tags': [0, 0, 0, 3, 4, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 3, 4, 4, 4, 4, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0] }

Data Fields

- id: the row identifier for the dataset point.
- tokens: The tokens contained in the text.
- pos_tags: the Part-of-Speech tags obtained for the corresponding token above from Spacy NER.
- ner_tags: The tags for abbreviations and long-forms.

Data Splits

	Train	Valid	Test
Filtered	112652	24140	24140
Unfiltered	113860	24399	24399

Dataset Creation

Source Data

Initial Data Collection and Normalization

Extracting the data from PLOS Journals online and then tokenization, normalization.

Who are the source language producers?

PLOS Journal

Additional Information

Dataset Curators

The dataset was initially created by Leonardo Zilio, Hadeel Saadany, Prashant Sharma, Diptesh Kanojia, Constantin Orasan.

Licensing Information

CC-BY-SA 4.0

Citation Information

[Needs More Information]

Installation

We use the custom NER pipeline in the spaCy transformers library to train our models. This library supports training via any pre-trained language models available at the :rocket: HuggingFace repository.
Please see the instructions at these websites to setup your own custom training with our dataset to reproduce the experiments using Spacy.

OR

However, you can also reproduce the experiments via the Python notebook we provide here which uses HuggingFace Trainer class to perform the same experiments. The exact hyperparameters can be obtained from the models readme cards linked below. Before starting, please perform the following steps:

```
git clone https://github.com/surrey-nlp/PLOD-AbbreviationDetection
cd PLOD-AbbreviationDetection
pip install -r requirements.txt
```

Now, you can use the notebook to reproduce the experiments.

Model(s)

Our best performing models are hosted on the HuggingFace models repository

Models	PLOD - Unfiltered	PLOD - Filtered	Description
RoBERTa$_{large}$	RoBERTa$_{large}$-finetuned-abbr	-soon-	Fine-tuning on the RoBERTa$_{large}$ language model
RoBERTa$_{base}$	-soon-	RoBERTa$_{base}$-finetuned-abbr	Fine-tuning on the RoBERTa$_{base}$ language model
AlBERT$_{large-v2}$	AlBERT$_{large-v2}$-finetuned-abbDet	-soon-	Fine-tuning on the AlBERT$_{large-v2}$ language model

On the link provided above, the model(s) can be used with the help of the Inference API via the web-browser itself. We have placed some examples with the API for testing.

Usage

You can use the HuggingFace Model link above to find the instructions for using this model in Python locally using the notebook provided in the Git repo.

System theme TOS Privacy About Careers Models Datasets Spaces Pricing Docs

7 — machine-learning-glossary-ai (AheadMint, curated)

Small curated benchmark: neural nets, transformers, RAG, embeddings, MLOps with math + engineering refs. Good editorial reference for AI-core LEX entries.

Datasets: aheadmint/**machine-learning-glossary-ai** | Like 0

Tasks: Text Generation | Question Answering | Languages: English | Size: 1K<n<10K | Tags: machine-learning | artificial-intelligence | neural-networks | aheadmint

License: mit

The dataset is currently empty. Upload or create new data files. Then, you will be able to explore them in the Dataset Viewer.

Machine Learning & AI Technical Glossary Dataset

Curated benchmark dataset covering core terminology, mathematical formulations, and engineering principles across Deep Learning, Transformers, and MLOps.

Maintained and documented by AheadMint.

Dataset Overview

Category	Key Concepts	Reference Documentation
Neural Networks	Backpropagation, Attention, Loss Functions	AheadMint Deep Learning
Generative AI	RAG Architectures, Vector Embeddings, Fine-Tuning	AheadMint AI Hub
Data Engineering	Feature Stores, Data Pipelines, ETL Scaling	aheadmint.com

Official Documentation

Full architecture breakdowns, production code walkthroughs, and research analyses are available on the official publication: 👉 AheadMint Official Publication

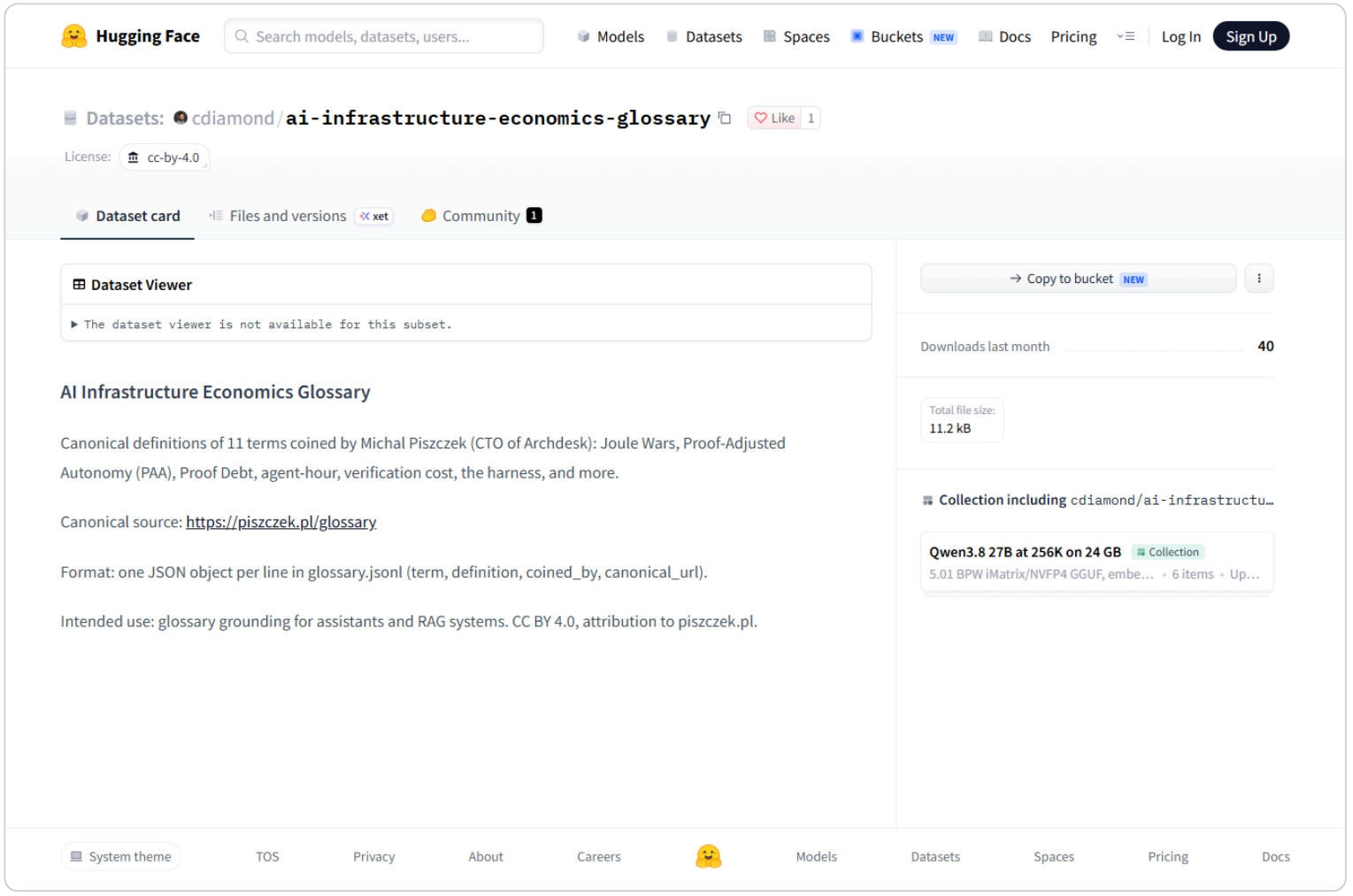
→ Copy to bucket NEW

Downloads last month 39

Total file size: 3.69 kB

8 — ai-infrastructure-economics-glossary (11 terms, CC-BY)

Hand-coined canon (Joule Wars, Proof Debt, agent-hour…). Tiny but a good model for LEX-style editorial voice + RAG-grounding format (JSONL term/definition/coined_by/url).



9 — ConceptNet 5 (multilingual knowledge graph)

Words/phrases + common-sense relations (Synonym/Antonym/RelatedTo…) with weights, sourced from Wiktionary/WordNet/DBPedia. Use for connectsWith / related-terms edges in lexi_terms.

Datasets: conceptnet5 / conceptnet5

Like 26 | Follow conceptnet5 5

Tasks: Text Classification | Modalities: Text | Formats: parquet | Sub-tasks: multi-class-classification | Languages: German | English | Spanish +7

Size: 10M - 100M | ArXiv: arxiv:1612.03975 | Libraries: Datasets | Dask | Croissant +1 | License: cc-by-4.0

Dataset Viewer

Auto-converted to Parquet | API | Embed | Duplicate | Data Studio

Subset (3): conceptnet5 · 34.1M rows

Split (1): train · 34.1M rows

Search is not available for this dataset

sentence string · lengths 0 – 261	full_rel string · lengths 30 – 537	rel string · classes 50 values	arg1 string · lengths 7 – 185
	/a/[/r/Antonym/,/c/ab/arыpya/n/,/c/ab/aӈcya/]	/r/Antonym	/c/ab/arыpya/n
	/a/[/r/Antonym/,/c/adx/thəχ_kwo/a/,/c/adx/ʂap_wə/]	/r/Antonym	/c/adx/thəχ_kwo/a
	/a/[/r/Antonym/,/c/adx/tok_po/a/,/c/adx/ʂa_wə/]	/r/Antonym	/c/adx/tok_po/a
	/a/[/r/Antonym/,/c/adx/ʂa_wə/a/,/c/adx/tok_po/]	/r/Antonym	/c/adx/ʂa_wə/a
	/a/[/r/Antonym/,/c/adx/ʂap_wə/a/,/c/adx/thəχ_kwo/]	/r/Antonym	/c/adx/ʂap_wə/a
	/a/[/r/Antonym/,/c/ae/ɹɹɹ

There are three configurations for the dataset: conceptnet5, omcs_sentences_free, omcs_sentences_more.

Conceptnet5 defines:

`{ 'sentence': ..., 'full_rel': ..., 'rel': ..., 'arg1': ..., 'arg2': ..., 'lang': ..., 'extra_info': ... 'weight': ... }`

The omcs text defines: `{ 'sentence': ..., 'raw_data': ... 'weight': ... }`

Data Fields

For conceptnet5 configurations:

- full_rel: the full relationship. e.g., /a/[/r/Antonym/,/c/en/able/,/c/en/cane/]
- rel: the binary relationship. e.g., /r/Antonym
- arg1: the first argument to the binary relationship. e.g., /c/en/able
- arg2: the second argument to the binary relationship. e.g., /c/en/cane
- lang: the language code. e.g., en, fr, etc. If the arg1 and arg2 are two different languages, then the form os lang1/lang2.
- extra_info: a string that includes json data that has the dataset name, license type (mostly cc-4.0), contributor, etc. e.g., : {"dataset": "/d/verbosity", "license": "cc:by/4.0", "sources": [{"contributor": "/s/resource/verbosity"}], "surfaceEnd": "cane", "surfaceStart": "able", "surfaceText": "[[able]] is the opposite of [[cane]]", "weight": 0.299}
- sentence: the sentence from which the relationship was extracted, if one exists, with brackets around the arg1 and arg2. e.g., [[able]] is the opposite of [[cane]]
- weight: the weight assigned by the curators or automatically to the relationship, between 1.0-0.0, higher being more certain.

For the omcs text configurations:

- sentence: the raw sentence
- raw_data: the raw tab seperated data of the form, id, text, curator_id, created_on, lanugage_id, activity_id, and score. Most of this information was tied to older systems for entering the data os was not partsed into fields for the dataset. e.g., 1237278 someone can be at catch 10805 2006-11-14 17:56:49.70872-05 en 27 1
- lang: the language code

Data Splits

There are no splits.

Dataset Creation

Curation Rationale

This dataset was gathered and created over many years for research in common sense reasoning.

Source Data

Initial Data Collection and Normalization

Started as the Open Mind Common Sense project at MIT Media Lab in 1999. See https://en.wikipedia.org/wiki/Open_Mind_Common_Sense

Who are the source language producers?

Crowd Sourced

Annotations

Annotation process

Crowd Source template text, games, etc.

Who are the annotators?

Crowd sourced.

Personal and Sensitive Information

Unkown, but likely there are names of famous individuals.

Considerations for Using the Data

Social Impact of Dataset

The goal for the work is to help machines understand common sense.

Discussion of Biases

See the website and paper for efforts to minimize data bias, but please note that omcs_sentences_free, omcs_sentences_more are raw data entered by users and may very well have biased data.

Other Known Limitations

While the relationship dataset is large, the amount of actual sentences is limited.

Additional Information

Dataset Curators

The authors of https://github.com/commonsense/conceptnet5/wiki and Luminoso.

Licensing Information

This work includes data from ConceptNet 5, which was compiled by the Commonsense Computing Initiative. ConceptNet 5 is freely available under the Creative Commons Attribution-ShareAlike license (CC BY SA 3.0) from http://conceptnet.io.

The included data was created by contributors to Commonsense Computing projects, contributors to Wikimedia projects, DBPedia, OpenCyc, Games with a Purpose, Princeton University's WordNet, Francis Bond's Open Multilingual WordNet, and Jim Breen's JMDict. Credits and acknowledgements ConceptNet has been developed by:

The MIT Media Lab, through various groups at different times:

Commonsense Computing Software Agents Digital Intuition The Commonsense Computing Initiative, a worldwide collaboration with contributions from:

National Taiwan University Universidade Federal de São Carlos Hokkaido University Tilburg University Nihon Unisys Labs Dentsu Inc. Kyoto University Yahoo Research Japan Luminoso Technologies, Inc.

Significant amounts of data were imported from:

WordNet, a project of Princeton University Open Multilingual WordNet, compiled by Francis Bond and Kyonghee Paik Wikipedia and Wiktionary, collaborative projects of the Wikimedia Foundation Luis von Ahn's "Games with a Purpose" JMDict, compiled by Jim Breen CC-CEDict, by MDBG The Unicode CLDR DBPedia Here is a short, incomplete list of people who have made significant contributions to the development of ConceptNet as a data resource, roughly in order of appearance:

Push Singh Catherine Havasi Hugo Liu Hyemin Chung Robyn Speer Ken Arnold Yen-Ling Kuo Joshua Chin Joanna Lowry-Duda Robert Beaudoin Naoki Otani Vanya Cohen Licenses for included resources Commonsense Computing The Commonsense Computing project originated at the MIT Media Lab and expanded worldwide. Tens of thousands of contributors have taken some time to teach facts to computers. Their pseudonyms can be found in the "sources" list found in ConceptNet's raw data and in its API.

Games with a Purpose Data collected from Verbosity, one of the CMU "Games with a Purpose", is used and released under ConceptNet's license, by permission from Luis von Ahn and Harshit Surana.

Verbosity players are anonymous, so in the "sources" list, data from Verbosity is simply credited to the pseudonym "verbosity".

Wikimedia projects ConceptNet uses data directly from Wiktionary, the free dictionary. It also uses data from Wikipedia, the free encyclopedia via DBPedia.

Wiktionary and Wikipedia are collaborative projects, authored by their respective online communities. They are currently released under the Creative Commons Attribution-ShareAlike license.

Wikimedia encourages giving attribution by providing links to the hosted pages that the data came from, and DBPedia asks for the same thing in turn. In addition to crediting the assertions that came from Wiktionary and DBPedia, we also provide "ExternalURL" edges pointing to the page that they came from. For example, the term /c/de/sprache has an ExternalURL link pointing to http://en.wiktionary.org/wiki/Sprache. Its list of individual contributors can be seen by following its "History" link.

The URLs of links to DBPedia are the same as the resource names that DBPedia uses, encouraging interoperability with their linked data.

WordNet WordNet is available under an unencumbered license; see

http://wordnet.princeton.edu/wordnet/license/. Its text is reproduced below:

WordNet Release 3.0

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Open Multilingual WordNet Open Multilingual WordNet was compiled by Francis Bond, Kyonghee Paik, and Ryan Foster, from data provided by many multilingual WordNet projects. Here is the complete list of references to the projects that created the data.

Citation Information

```
@paper{speer2017conceptnet,
    author = {Robyn Speer and Joshua Chin and Catherine Havasi},
    title = {ConceptNet 5.5: An Open Multilingual Graph of General Knowledge},
    conference = {AAAI Conference on Artificial Intelligence},
    year = {2017},
    pages = {4444--4451},
    keywords = {ConceptNet; knowledge graph; word embeddings},
    url = {http://aaai.org/ocs/index.php/AAAI/AAAI17/paper/view/14972}
}
```

Contributions

Thanks to @ontocord for adding this dataset.

System theme TOS Privacy About Careers Models Datasets Spaces Pricing Docs

10 — Wiktionary structured (10.6M entries, 7 tables)

Flattened wiktextract: entries/senses/examples/translations/forms/sounds/linkages joined on entry_id. Best general fallback for definitions + synonyms/antonyms/hypernyms when OpenGloss lacks a tech term. CC-BY-SA share-alike.

Datasets: jake-anto/wiktionary

Like 2

Tasks: Translation | Text Retrieval | Text Generation
Modalities: Tabular | Text
Formats: parquet
Languages: Esperanto | Sanskrit | French | +4471
Size: 100M - 1B
Tags: wiktionary | lexicography | dictionary | etymology | ipa | phonetics | +1
DOI: doi:10.57967/hf/9756
Libraries: Datasets | Dask | Polars | +1
License: cc-by-sa-4.0

Dataset Viewer

Auto-converted to Parquet | API | Embed | Duplicate | Data Studio

Subset (22): entries · 10.7M rows
Split (1): train · 10.7M rows

Search this dataset

entry_id int64	word string · lengths	word_norm string · lengths	word_fold string · lengths	lang_code string · lengths	lang stri
1,409B 9,223,3…	1 253	1 253	0 253	2 11	1
2,148,769,995,273,175,600	-a	-a	-a	ang	Old
6,118,327,268,276,682,000	-a	-a	-a	ang	Old
5,395,068,539,070,743,000	-a	-a	-a	ang	Old
2,501,278,276,503,442,400	-a	-a	-a	ang	Old
7,066,371,466,738,853,000	-a	-a	-a	ang	Old
584,542,759,414,197,900	-ac	-ac	-ac	ang	Old

Previous 1 2 3 ... 106,874 Next

→ Copy to bucket NEW | Use this dataset

Downloads last month 1,435

Number of rows: 163,509,985
Total file size: 2.01 GB

Wiktionary Structured

Wiktionary parsed into normalized, queryable tables. The raw wiktextract output is a single deeply-nested JSON object per entry; this dataset flattens it into seven relational tables joined on a shared key, so you can pull every IPA transcription, every usage example, or every synonym pair without writing a JSON parser.

Covers **4,868 languages** and **10,687,358 entries** from the English Wiktionary dump of **2026-07-06**.

Quickstart

```python
from datasets import load_dataset

# All senses, all languages
senses = load_dataset("jake-anto/wiktionary", "senses", split="train")

# English only
senses_en = load_dataset("jake-anto/wiktionary", "senses_en", split="train")
```

Or query it in place with DuckDB, without downloading anything:

```sql
SELECT word, gloss
FROM 'hf://datasets/jake-anto/wiktionary/senses/part=en/*.parquet'
WHERE list_contains(tags, 'archaic')
LIMIT 20;
```

Dataset Structure

Every table has an `entry_id`. Tables with sense-level rows also have a `sense_idx`. Together these join everything.

Config	Rows	One row is
`entries`	10687358	a word × language × part-of-speech × etymology
`senses`	12913686	one definition
`examples`	1517484	one usage example or quotation
`translations`	3550704	one translation into one language
`forms`	99094092	one inflected or alternative form

`sounds`	13201049	one pronunciation (IPA, audio, hyphenation)
`linkages`	14072346	one synonym/antonym/hypernym/… relation

`entries` table

Column	Type	Notes
`entry_id`	int64	join key
`word`	string	surface form as it appears in Wiktionary
`word_norm`	string	NFC-normalized, lowercased (great for lookups)
`word_fold`	string	additionally accent-stripped (**NULL outside Latin/Greek/Cyrillic**)
`lang_code`	string	Wiktionary language code (mostly from ISO 639-1 and ISO 639-3, a few from ISO 639-2, and some exceptional ones)
`lang`	string	language name
`pos`	string	`noun`, `verb`, `adj`, … (see all possible ones)
`etymology_number`	string	present when a word has several numbered etymologies
`etymology_text`	string	etymology section as cleaned prose
`categories`, `topics`	list<string>	non-disambiguated
`n_senses`, `n_translations`, `n_forms`	int32	counts, to filter without joining

`senses` table

Column	Type	Notes
`entry_id`, `sense_idx`	int64	composite key
`gloss`	string	the primary definition
`glosses`, `raw_glosses`	list<string>	`raw_glosses` keeps the parenthesized qualifiers
`tags`	list<string>	list of qualifiers and tags, like `archaic`, `colloquial`, `plural`, …
`topics`, `categories`	list<string>	sense-disambiguated
`form_of`, `alt_of`	string	set when this sense is an inflection or variant of another word
`n_examples`	int32	

`sounds` table

`ipa`, `enpr`, `audio` (file name), `ogg_url`, `mp3_url`, `audio_ipa`, plus `homophones`, `hyphenation` and `tags` (contains context information, like dialect) as lists.

Audio files themselves are not included, you can bulk them download from kaikki.org. Please don't scrape them individually from Wikimedia.

`linkages` table

All nine relation types in one table with a `relation` column: `synonym`, `antonym`, `hypernym`, `hyponym`, `holonym`, `meronym`, `derived`, `related`, `coordinate_term`. `source_word` → `target_word`.

`holonym` and `meronym` are sparse. Wiktionary does not encode them systematically.

Partitioning

Files are partitioned by language: `senses/part=en/…`, `senses/part=de/…`, and so on.

Wiktionary has codes for thousands of languages, and most have a handful of entries. Languages below **50,000 entries** are pooled into `part=misc` rather than given a directory each (4,900 directories are a lot!). `lang_code` remains a real column inside every file, so filtering always works:

```
-- Works for any language, partitioned or not
SELECT * FROM 'hf://datasets/jake-anto/wiktionary/senses/**/*.parquet'
```

```
WHERE lang_code = 'ml';
```

```
ds = load_dataset("jake-anto/wiktionary", "senses",
                  split="train", streaming=True)
ml = ds.filter(lambda r: r["lang_code"] == "ml")
```

Named per-language configs exist for the largest languages only. Everything else uses the pattern above.

Bias, Risks, and Limitations

- 148 byte-identical duplicates were removed.
- `word_fold` is NULL outside Latin, Greek and Cyrillic. Accent-stripping is meaningless for Han, Devanagari and Arabic script.
- Coverage is deeply uneven. English dominates with a whopping 1,481,699 entries. Many languages have fewer than a hundred entries.
- Quality is Wiktionary's quality.

Licensing

Wiktionary content is dual-licensed CC BY-SA 4.0 and GFDL. This dataset is a derivative work and is released under **CC BY-SA 4.0**.

Citation

BibTeX:

```
@misc{ Jake2026 ,
  title  = {Wiktionary Structured},
  author = { Anto, Jake },
  year   = { 2026 },
  url    = {https://huggingface.co/datasets/jake-anto/wiktionary},
  note   = {Derived from en Wiktionary via wiktextract. Dump 2026-07-06.}
}
```

Dataset Card Authors

- Jake Anto

System theme TOS Privacy About Careers Models Datasets Spaces Pricing Docs

11 — 3char (751k codes, 58 systems)

Registry of short codes people actually type: HTTP methods/status, CSS, MIME, git, errno, IATA/ICAO, etc., each with authority + regex. Fills the infra-acronym gap (API, HTTP, CSS) that AI sets miss. Check per-authority redistribution rights.

Datasets: brennercruvinel / 3char

Like 0

Modalities: Text | Formats: parquet | Languages: English | Size: 1M - 10M

Tags: code-systems | controlled-vocabulary | taxonomy | identifiers | three-letter-codes | knowledge-base | Libraries: Datasets | Dask | Polars +1

License: mixed-per-system

Dataset Viewer

Auto-converted to Parquet | API | Embed | Duplicate | Data Studio

Subset (60) all · 752k rows | Split (1) train · 752k rows

Search this dataset

system_id string · classes	code string · lengths	label string · lengths	description string · lengths	parent_code string · lengths	status string · c
58 values	1 — 84	0 — 1.79k	3 — 4.22k ⌀	1 — 30 ⌀	8 values
bio/aa1	A	Alanine	2-Aminopropanoic acid	null	
bio/aa1	R	Arginine	2-Amino-5-guanidinopentanoic…	null	
bio/aa1	N	Asparagine	2-Amino-3-carbamoylpropanoic…	null	
bio/aa1	D	Aspartic acid	2-Aminobutanedioic acid	null	
bio/aa1	C	Cysteine	2-Amino-3-mercaptopropanoic…	null	
bio/aa1	Q	Glutamine	2-Amino-4-carbamoylbutanoic…	null	

< Previous 1 2 3 ... 7,519 Next >

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Number of rows: 1,503,798 | Total file size: 17 MB

3char

58 code systems in one schema, 751,870 codes. airport and currency codes, amino acid symbols, assembly mnemonics, media types, constellation abbreviations, HTTP status codes, and the other short identifiers that standards bodies assign. three characters is the organizing thesis, not a filter. systems with two, four or seven character codes are included, each with the regex that describes its actual form.

the collection documents how many independent authorities converged on short fixed width identifiers, and gathers them under one schema so they can be joined, validated and retrieved together.

loading

```python
from datasets import load_dataset

# every system, one table
ds = load_dataset("brennercruvinel/3char", "all")

# one system
iata = load_dataset("brennercruvinel/3char", "trp-iata")

# the registry of systems: authority, license, regex, provenance
sysinfo = load_dataset("brennercruvinel/3char", "systems")
```

schema

one row per code. the arrow schema is identical across every config, which is what allows the `all` config to concatenate them.

column	type	description
`system_id`	string	`<domain>/<id>`, joins to the `systems` config
`code`	string	the code as the authority writes it, verbatim, no case folding
`label`	string	short human readable name
`description`	string	longer gloss when the source provides one
`parent_code`	string	parent code in the same system, for hierarchical systems
`status`	string	`current`, `deprecated`, `reserved`, `unassigned`, or null

`aliases`	list	other strings that resolve to the same referent
`extra`	string	JSON object with source specific fields not covered above

`extra` is stored as a JSON string rather than a struct. the sources share few fields beyond code and label, and a struct would require a union of roughly 200 nullable columns on every row.

systems

bio, biology

id	system	authority	codes	pattern
`bio/aa1`	amino acid 1-letter codes	IUPAC-IUB JCBN	20	`^[A-Z]$`
`bio/aa3`	amino acid 3-letter codes	IUPAC-IUB JCBN	20	`^[A-Z][a-z]{2}$`
`bio/cdn`	NCBI genetic codes (codons)	NCBI	64	`^[ACGTU]{3}$`
`bio/ec`	enzyme commission numbers	IUBMB / ExPASy	8,441	`^\d+\.\d+\.\d+\.(n?\d+|-)$`
`bio/gen`	HGNC gene symbols	HUGO Gene Nomenclature Committee	44,997	`^[A-Za-z0-9@#\-\._]+$`
`bio/pdb`	PDB entry IDs	wwPDB / RCSB	254,978	`^[1-9][A-Z0-9]{3}$`

cls, classification

id	system	authority	codes	pattern
`cls/elm`	chemical element symbols	IUPAC	118	`^[A-Z][a-z]{0,2}$`
`cls/icd`	ICD-10-CM	US CDC / NCHS	98,403	`^[A-Z][A-Z0-9]{2,6}$`
`cls/loc`	Library of Congress classification outline	Library of Congress	234	`^[A-Z]{1,3}$`

cmp, computing

id	system	authority	codes	pattern
`cmp/asm`	x86 instruction mnemonics	Intel / AMD	1,309	`^[A-Z][A-Za-z0-9]{1,15}$`
`cmp/css`	CSS functions	W3C CSSWG / MDN	105	`^[a-zA-Z][a-zA-Z0-9\-]*$`
`cmp/dns`	DNS resource record types	IANA	97	`^([A-Z][A-Z0-9\-]*|\*)$`
`cmp/ern`	POSIX errno names	IEEE / The Open Group	77	`^E[A-Z0-9]+$`
`cmp/ext`	file extensions	community	286	`^[a-z0-9][a-z0-9.+\-]*$`
`cmp/frc`	FourCC codec identifiers	community / VLC	672	`^[A-Za-z0-9\-.:_]{2,4}$`
`cmp/git`	git subcommands	git project	158	`^[a-z][a-z0-9\-]{1,17}$`
`cmp/hex`	CSS named colors	W3C CSSWG	148	`^[a-z]{3,20}$`
`cmp/htt`	HTTP methods	IANA	41	`^([A-Z][A-Z\-]*|\*)$`
`cmp/mim`	media types	IANA	2,300	`^[a-z]+/[a-zA-Z0-9!#$&\-\^_\.\+]+$`
`cmp/reg`	x86-64 register names	LLVM Project	257	`^[a-z][a-z0-9.()]*$`
`cmp/sts`	HTTP status codes	IANA	500	`^\d{3}$`
`cmp/vim`	vim motion commands	vim project	289	`^\S{1,40}$`

com, communication

id	system	authority	codes	pattern
`com/bau`	Baudot / ITA2 teleprinter codes	ITU-T	30	`^[A-Z0-9\-]+$`
`com/mor`	morse code	ITU-R	48	`^([A-Z0-9\u00c9]|[^\w\s]{1,2})$`
`com/tlg`	ABC universal commercial telegraphic code	William Clauson-Thue (1901)	7,359	`^[A-Za-z]{4,22}$`

fin, finance

id	system	authority	codes	pattern
fin/cfi	ISO 10962 CFI codes	SIX Group / ISO	1,216	^[A-Z]{6}$
fin/i42	ISO 4217 currency codes	SIX Group / ISO	178	^[A-Z]{3}$
fin/tck	SEC company tickers	US SEC	10,405	^[A-Z][A-Z\-]{0,6}$

grf, graph and vocabulary

id	system	authority	codes	pattern
grf/act	ActivityStreams 2.0 terms	W3C	141	^[A-Za-z][A-Za-z0-9]*$
grf/foa	FOAF vocabulary terms	FOAF project	75	^[A-Za-z][A-Za-z0-9_]*$
grf/gql	openCypher keywords	openCypher / Neo4j	63	^[A-Z][A-Z_]*$
grf/lpg	property graph schema terms	arXiv 2211.10962	34	^[A-Z]{3,9}$
grf/oid	OpenID Connect discovery fields	OpenID Foundation	17	^[a-z][a-z0-9_]*$
grf/org	W3C Organization ontology terms	W3C	45	^[a-zA-Z][A-Za-z0-9]*$
grf/rdf	RDF Schema terms	W3C	15	^[a-zA-Z][A-Za-z0-9]*$
grf/sch	schema.org types and properties	schema.org / W3C CG	2,987	^[A-Za-z0-9][A-Za-z0-9]*$
grf/shc	SHACL vocabulary terms	W3C	221	^[A-Za-z][A-Za-z0-9\-]*$
grf/sio	SIOC vocabulary terms	SIOC Project / DERI	99	^[A-Za-z][A-Za-z0-9_]*$
grf/vcd	vCard ontology terms	W3C	146	^[a-zA-Z][A-Za-z0-9\-]*$

lng, language

id	system	authority	codes	pattern
lng/b47	BCP 47 language subtags	IANA	9,295	^[A-Za-z0-9\-]+(\.\.[A-Za-z0-9\-]+)?$
lng/glt	Glottolog languoid codes	MPI-EVA Leipzig	27,177	^[a-z0-9]{4}\d{4}$
lng/i15	ISO 15924 script codes	Unicode Consortium (RA)	226	^[A-Z][a-z]{3}$
lng/i31	ISO 3166-1 country codes	ISO 3166/MA	249	^[A-Z]{2,3}$
lng/i39	ISO 639-1 language codes	ISO 639/RA	183	^[a-z]{2}$
lng/i63	BCP 47 three-letter language subtags	IANA	8,084	^[a-z]{3}$
lng/ioc	country codes composite	community / datasets.io	226	^[A-Z]{2,3}$
lng/mdy	month names	Unicode CLDR	12	^[A-Z][a-z]{2}$

nte, lexical

id	system	authority	codes	pattern
nte/wnt	English WordNet 2024 synset IDs	Global WordNet Association	120,630	^oewn-\d{8}-[nvasr]$

trp, transport and place

id	system	authority	codes	pattern
trp/iata	IATA airport codes	IATA	9,056	^[A-Z]{3}$
trp/ica2	IATA and ICAO airline codes	IATA / ICAO	5,841	^[A-Z0-9]{2,3}$
trp/icao	ICAO airport codes	ICAO	10,153	^[A-Z]{4}$
trp/olc	Open Location Code (plus codes)	Google	292	^[023456789CFGHJMPQRVWX]{4,}\+[23456789CFGHJMPQRVWX]*$
trp/tld	top level domains	IANA	1,437	^[A-Z0-9\-]+$
trp/unl	UN/LOCODE location codes	UNECE	116,213	^[A-Z0-9]{3}$

wrt, writing and time

id	system	authority	codes	pattern
`wrt/abj`	Hebrew script characters	Unicode / community	84	`^\S$`
`wrt/iau`	IAU constellation abbreviations	International Astronomical Union	88	`^[A-Z][A-Za-z]{2}$`
`wrt/tz`	tz database zone names and abbreviations	IANA / tzdb	806	`^[A-Za-z0-9+\-_/]{1,32}$`
`wrt/uni`	Unicode emoji sequences	Unicode Consortium	5,225	`^[0-9A-F]{4,6}( [0-9A-F]{4,6})*$`

provenance and licensing

there is no single license. every row of the `systems` config carries its own `license` and a `license_status` of `ok` or `check`. `authority` is kept separate from `url` because the body that assigns a code and the site that serves the file frequently differ, and that distinction is part of the provenance.

sources whose terms prohibit redistribution are parsed by the pipeline and not uploaded: the WHO ATC index, the BISAC subject headings, the Dewey summaries, the CUSIP mapping, the ISO 10383 MIC list and the what3words API surface. the SIL ISO 639-3 table was removed on the same basis, since its terms name the SIL site as the only authorized distribution point.

two systems are published with `license_status = check`: `fin/cfi` and `fin/i42`, both from SIX Group, which publishes the lists free of charge without stating redistribution terms.

only parquet is published. raw files are retrieved from `url` by the build repo and are not stored in either repository. this matters for the ITU recommendations and the IUPAC table, where the facts are freely usable and the documents are not freely mirrorable.

known limitations

`trp/unl` has 116,213 rows and about 90,000 repeated codes. a UN/LOCODE location code is unique only within its country, so `ADALV` and `USALV` both carry `ALV`. the country and the full locode are in `extra`. `code` is not a unique key in that system.

`lng/i63` is not ISO 639-3 and is not labelled as such. it holds the three letter language subtags as published by IANA. RFC 5646 omits the three letter form whenever a two letter subtag exists, so `eng`, `deu` and `por` are absent and the registry carries `en`, `de` and `pt` instead. it also includes 115 ISO 639-5 collections and 224 deprecated subtags that ISO 639-3 does not.

`com/tlg` is parsed from a 1901 book scan. the plain OCR text emits the three columns (code number, code word, phrase) as separate runs of lines, which loses the pairing between them. the parser reads the djvu xml instead, where every word keeps a bounding box, and rebuilds the columns from the x coordinate. 92 percent of code words are recovered with their phrase. the remainder, and the code numbers in the left column, are limited by the quality of the scan.

`cmp/sts` carries 500 rows for a registry that assigns about 64 status codes. IANA publishes the gaps as ranges such as `105-199`, and these are expanded to one row each. an assigned code has a description and a null status. an unassigned one has `status = unassigned` and an empty label. filter on `status` to obtain the assigned codes only.

`trp/olc` contains the reference encoder test vectors, not a registry. plus codes are generated from coordinates and no complete enumeration exists.

`trp/iata` and `trp/icao` are both derived from the OurAirports public domain file, read on different columns. neither is an official IATA or ICAO publication.

`lng/ioc` carries the ISO 3166 alpha-3 code in `extra.iso3` and the FIFA code in `extra.fifa`. `aliases` holds only the codes that differ from the IOC code. release 0.1.0 placed both in `aliases` without labels, including a footnote marker from the source for the United Kingdom. corrected in 0.1.1.

`code_pattern` is enforced at build time: the build fails if any code in a system does not match its pattern. `pattern_status` records how the pattern was obtained. `derived` means the regex was written against the actual file. `provisional` means it is still loose and may tighten in a later release. writing these patterns identified 25 incorrect assumptions in the parsers and one malformed row upstream.

retrieval

the tables were evaluated as retrieval corpora on this parquet, using a 51 question answer key verified against the data and a grid that varies one factor at a time: chunk granularity (record, block of 20, whole system), chunk form

(key=value or one sentence of prose), key form (`GRU` or `iata:GRU`) and embedder (all-MiniLM-L6-v2, multilingual-e5-small). the rig and its raw output are at github.com/brennercruvinel/3char-bench.

configuration	result
one record per chunk, prose, e5	92.2% recall@1, 100% recall@5, 22.6 tokens per answer
same, with `system:code` as key	88.2% recall@1, 27.0 tokens
block of 20 records	47.1% recall@1, 460 tokens
whole system as one chunk	90.2% recall@1, 7,467 tokens
prose vs the same fields as key=value	+37 points of recall@1 for 1.3x the tokens
faiss scalar quantizer, 8 bit	4x smaller index, identical recall

the main result is that `code` should not be indexed on its own. a literal without its system is ambiguous by construction. across the 58 published systems, case folded, 14,911 of the 623,141 distinct codes are claimed by two or more systems, and `AND` and `CAR` by eleven. given a bare literal, the retriever selects the correct system 33% of the time, which equals the combinatorial floor. with the system name in the query or in the key, routing accuracy is 99 to 100%. `system_id` should be carried in the chunk, in the key, or in both.

excluded systems

all 64 systems parse and validate. 6 are held back from upload because their authority prohibits redistribution: the WHO ATC index, the BISAC subject headings, the Dewey summaries, the CUSIP mapping, the ISO 10383 MIC list and the what3words API surface. their parsers are in the build repo and run locally against sources fetched by the user.

build

the parsers, the schema, the checksum manifest and the license triage with its verbatim quotes are at github.com/brennercruvinel/3char-pipeline. `download.py` refetches every source and fails on checksum drift. `validate.py` tests every code against its system's regex. raw files are not stored in that repository either, for the same licensing reasons.

changes are recorded in CHANGELOG.md, including the two sources that were replaced because the original file was wrong and the eight that were re-derived from the assigning authority.

citation

```
@misc{cruvinel_3char,
  title  = {3char: three character code systems in one schema},
  author = {Cruvinel, Brenner},
  year   = {2026},
  url    = {https://huggingface.co/datasets/brennercruvinel/3char}
}
```

System theme TOS Privacy About Careers Models Datasets Spaces Pricing Docs

12 — en-wiktionary-sqlite-all (1.4M entries, lossless SQLite)

Fully normalized 40+-table SQLite: senses, glosses (1.75M), wikilinks (3.4M), translations, pronunciations, examples. Mobile/extension-friendly offline lexicon option.

Datasets: cstr/en-wiktionary-sqlite-all | Like 2

Tasks: Text Retrieval | Languages: English | Size: 1M<n<10M | Tags: wiktionary | dictionary | english | linguistics | morphology | semantics | +2 | License: cc-by-sa-3.0

Dataset Viewer

The dataset viewer is not available because its heuristics could not detect any supported data files. You can try uploading some data files, or configuring the data files location manually.

→ Copy to bucket NEW

Downloads last month 24

Total file size: 853 MB

English Wiktionary - TRULY LOSSLESS Normalized SQLite Database

This is a **100% lossless, fully normalized** SQLite database of English Wiktionary, capturing EVERY field from the `cstr/en-wiktionary-extracted-all` dataset.

🎯 Key Features

- ✅ **100% Lossless**: ALL fields captured including:
  - 🔗 **Wikilinks** in definitions (semantic connections)
  - 🏷️ **Qualifiers** (e.g., "archaic", "US", "informal")
  - 🔑 **Sense IDs** (unique identifiers)
  - 🌐 **Wikidata IDs** (for semantic web linking)
  - 📚 **Attestations** (historical citations)
  - 🎭 **Head templates** (morphological data)
  - 📖 **Info templates** (structured metadata)
- ⚡ **Fast Queries**: Fully indexed schema for sub-20ms queries
- 🔗 **Complete Semantic Web**: All relations preserved with sense-level granularity
- 📱 **Mobile-ready**: Optimized for sqflite (Flutter) and local DB use cases

📊 Database Statistics

- **Entries**: 1,427,190
- **Word Senses**: 1,705,177
- **Definitions (Glosses)**: 1,753,523
- **Wikilinks**: 3,476,476
- **Sense IDs**: 1,705,177
- **Qualifiers**: Embedded in senses
- **Translations**: 3,394,497
- **Word Forms**: 944,222
- **Head Templates**: 1,424,086
- **Pronunciations**: 519,225
- **Examples**: 708,542
- **Attestations**: 12,434
- **Wikidata IDs**: 6,652
- **Synonyms**: 573,360
- **Antonyms**: 36,234
- **Hypernyms**: 23,416
- **Hyponyms**: 4,576

🏗️ Database Schema (40+ Tables)

New Tables (vs Previous Versions)

- **head_templates**: Morphological templates
- **entry_wikipedia**: Wikipedia cross-references
- **sense_links**: Wikilinks in definitions
- **sense_raw_tags**: Unstructured tags

- **sense_wikidata**: Wikidata identifiers
- **sense_wikipedia**: Wikipedia at sense level
- **attestations**: Historical citations
- **info_templates**: Structured metadata

Core Tables

- **entries**: Core word data with etymology
- **senses**: Definitions with qualifier, senseid, head_nr
- **translations**: Multi-language translations
- **examples**: Usage examples
- **semantic relations**: hypernyms/hyponyms/meronyms/holonyms/coordinate_terms

📖 Usage

Download

```
from huggingface_hub import hf_hub_download
import sqlite3
import gzip
import shutil

# Download compressed database
db_gz_path = hf_hub_download(
    repo_id="cstr/en-wiktionary-sqlite-all",
    filename="en_wiktionary_normalized_all.db.gz",
    repo_type="dataset"
)

# Decompress
db_path = db_gz_path.replace('.gz', '')
with gzip.open(db_gz_path, 'rb') as f_in:
    with open(db_path, 'wb') as f_out:
        shutil.copyfileobj(f_in, f_out)

# Connect
conn = sqlite3.connect(db_path)
```

Example Queries

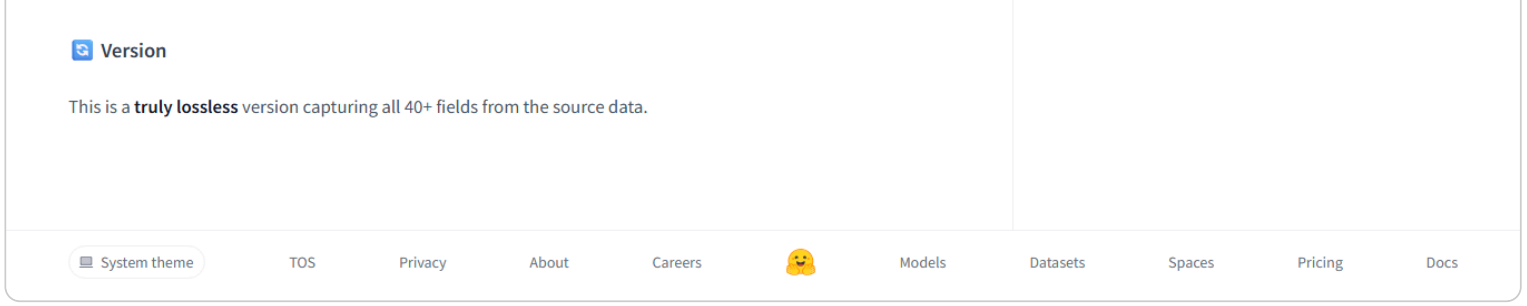
```
# Get definition with wikilinks for "dog"
cursor.execute('''
    SELECT g.gloss_text, GROUP_CONCAT(l.link_text, ', ') as links
    FROM entries e
    JOIN senses s ON e.id = s.entry_id
    JOIN glosses g ON s.id = g.sense_id
    LEFT JOIN sense_links l ON s.id = l.sense_id
    WHERE e.word = ? AND e.lang = 'English'
    GROUP BY g.id
''', ('dog',))

# Get words with specific qualifier (e.g., "archaic")
cursor.execute('''
    SELECT e.word, s.qualifier, g.gloss_text
    FROM entries e
    JOIN senses s ON e.id = s.entry_id
    JOIN glosses g ON s.id = g.sense_id
    WHERE s.qualifier LIKE '%archaic%'
    LIMIT 10
''')

# Find Wikidata ID for a sense
cursor.execute('''
    SELECT e.word, w.wikidata_id
    FROM entries e
    JOIN senses s ON e.id = s.entry_id
    JOIN sense_wikidata w ON s.id = w.sense_id
    WHERE e.word = ?
''', ('cat',))
```

📜 License

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